Finite Groups and Their Representations — Exercise Sheet 1

Groups acting

Variant: Questions only

Exercise 1 (An action with a non-trivial kernel). Let $G = C_{12} = \langle \zeta \rangle$, where $\zeta = \exp(2\pi \mathrm{i}/12)$, and let $X = \{0, 1, \ldots, 7\}$. For $j \in \mathbb{Z}$ and $x \in X$, define $\zeta^j \cdot x$ to be the unique element of X congruent to $x + 2j$ modulo 8.

(i) Prove that the rule is independent of the exponent chosen for ζ^j, and that it defines a left action.

(ii) Find all orbits and the stabiliser of every point. Verify orbit–stabiliser numerically on each orbit.

(iii) Find the action's kernel. Write the permutation of X induced by ζ, and determine the order of the image in S_X.

Exercise 2 (A smallest faithful action of C_6). Let $G = C_6 = \langle \zeta \rangle$, where $\zeta = \exp(2\pi \mathrm{i}/6)$.

(i) Prove that sending ζ to $(1\,2\,3)(4\,5)$ gives a faithful action on $\{1, 2, 3, 4, 5\}$. Find its orbits and stabilisers.

(ii) Prove that G has no faithful action on a non-empty set with fewer than five points. Do not use the general cycle-type results of Lecture 2.

Exercise 3 (Groups of order four). In S_4, put

$$a = (1\,2)(3\,4), \qquad b = (1\,3)(2\,4), \qquad c = (1\,4)(2\,3), \qquad V_4 = \{e, a, b, c\}.$$

(i) Show that V_4 is a subgroup and write its multiplication table. This subgroup is called the *Klein four-group*.

(ii) Prove that every group of order 4 is isomorphic to C_4 or V_4. Construct an isomorphism in each case.

(iii) Prove that these two groups are not isomorphic.

Exercise 4 (Conjugacy in the dihedral groups). Let $n \geq 3$. Use the normal form from Lecture 1:

$$D_n = \{r^a, r^a s : 0 \leq a < n\}, \qquad r^n = s^2 = e, \qquad srs^{-1} = r^{-1}.$$

(i) Compute conjugation of a rotation by a rotation and by a reflection. Compute conjugation of $r^b s$ by r^a and by s.

(ii) Determine all conjugacy classes, the centre, and the centraliser of each element. Distinguish odd and even n.

(iii) For D_5 and D_6, give a table with one representative from every class, its class size and its centraliser. Write the class equation in each case.

Exercise 5 (Necklaces and bracelets with three black beads). A regular heptagon has vertices $1, \dots, 7$, in cyclic order. Exactly three vertices are black and the other four are white. Let X be the set of the resulting colourings, identified with their three-element sets of black vertices. Put

$$r = (1\,2\,3\,4\,5\,6\,7), \qquad s = (2\,7)(3\,6)(4\,5).$$

The group $D_7 = \langle r, s \rangle$ acts by $g \cdot A = \{g(a) : a \in A\}$. A *necklace* is an orbit under $\langle r \rangle$; a *bracelet* is an orbit under D_7.

(i) Count the colourings fixed by each rotation and each reflection.

(ii) Use orbit counting to find the numbers of necklaces and bracelets.

(iii) Give one representative of each necklace. Identify the distinct necklaces that become the same bracelet.

Exercise 6 (Unique square roots). Let G be a finite group, including the possibility $G = \{e\}$. The squaring map is the set map $q \colon G \to G$ given by $q(g) = g^2$. It is not assumed to be a homomorphism.

(i) Prove that q is bijective if and only if $|G|$ is odd.

(ii) When $m = |G|$ is odd, give the unique square root of every $g \in G$ as a power of g. In particular, give it when $m = 15$.

Exercise 7 ⋆. Let G be the group of rotations of the cube $[-1, 1]^3$. Here a rotation means an orthogonal linear map of $\mathbb{R}^3$ with determinant 1 that preserves the cube. Lecture 1's self-check found $|G| = 24$ by orbit–stabiliser on the faces; take this count as given. The four body-diagonal lines are $L_i = \mathbb{R}v_i$, where

$$v_1 = (1, 1, 1), \quad v_2 = (1, -1, -1), \quad v_3 = (-1, 1, -1), \quad v_4 = (-1, -1, 1).$$

(i) Show that every element of G permutes the four lines, giving a homomorphism $\rho \colon G \to S_4$.

(ii) Prove that this action is faithful. A rotation that fixes a line is allowed to interchange its two endpoints, so fixing the four lines does not immediately mean fixing the eight vertices.

(iii) Deduce that $G \cong S_4$.